

Real-Time Graph Neural Network Transaction Monitoring

1. Problem Statement

Traditional Anti-Money Laundering (AML) systems mainly depend on predefined rules and isolated transaction analysis. These approaches may fail to identify complex relationships between multiple accounts, transactions, and suspicious financial networks. Money laundering often involves several accounts, repeated transfers, circular transactions, and coordinated activities that are difficult to detect using conventional methods.

The proposed project develops a **Real-Time Graph Neural Network (GNN)-based AML transaction monitoring system** that represents bank accounts as nodes and financial transactions as edges. The GNN analyzes relationships and transaction patterns to identify potentially suspicious activities. An **MLOps and CI/CD pipeline** is incorporated to continuously monitor model performance, automate retraining, testing, validation, and deployment, helping maintain reliable detection performance over time.

2. Abstract

Money laundering is a major financial crime in which illegal funds are transferred through multiple accounts and transactions to hide their original source. Conventional AML systems generally rely on predefined rules and individual transaction analysis, making it difficult to detect sophisticated and interconnected money-laundering patterns.

This project proposes a **Real-Time Graph Neural Network (GNN)-based Anti-Money Laundering transaction monitoring system**. Bank accounts are represented as nodes and financial transactions as edges, forming a transaction graph. A GNN model analyzes the structural relationships and transaction features to identify suspicious activities such as circular transactions, unusual fund transfers, and interconnected account networks. The system processes transactions in real time and generates risk scores or alerts for potentially suspicious activities.

To ensure continuous model reliability, an **MLOps and CI/CD pipeline** is integrated for automated model testing, performance monitoring, retraining, model versioning, validation, and deployment. The proposed system aims to improve detection of complex financial crime patterns while reducing dependency on manually defined rules.

3. Advantages

- Detects complex relationships between multiple accounts and transactions.
- Supports real-time monitoring of incoming financial transactions.
- Can identify suspicious patterns that rule-based systems may miss.
- Provides better visibility into interconnected transaction networks.
- Enables automated model performance monitoring and retraining through MLOps.
- Supports automated testing and model deployment through CI/CD.
- Reduces manual effort through automated risk scoring and alert generation.

- Can adapt to changing transaction patterns through continuous model updates.
- Supports centralized monitoring through an AML investigation dashboard.
- Combines GNN, real-time processing, MLOps, and CI/CD in one architecture.

4. Disadvantages

- Large transaction graphs can require significant memory and computational resources.
- GNN-based systems are more complex to design and implement than traditional ML models.
- Detection performance depends strongly on the quality and availability of transaction data.
- The system may generate false positives for legitimate transactions.
- Training and updating large graph models can be time-consuming.
- GNN decisions can be difficult to explain clearly to investigators.
- Financial transaction data requires strong privacy, security, and access controls.
- Real-time streaming, graph databases, model serving, and monitoring can increase infrastructure costs.
- Changes in criminal behavior can cause concept drift and reduce model performance.
- The MLOps pipeline requires continuous maintenance and monitoring.

Working Flow

Transaction Input → Data Preprocessing → Graph Construction → Real-Time Processing → GNN Analysis → Risk Score → AML Alert → Investigation → Performance Monitoring → Automatic Retraining → CI/CD Validation → Model Deployment